DOMESTIC AIRLINES — TRAINING

CPDLC — Controller-Pilot Data Link

Course handbook

Controller-pilot data link communications: references, data link services, message principles, phraseology, failures and the B737 CMU-900.

6 lessons · 2 modules · about 0.5 hours

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.

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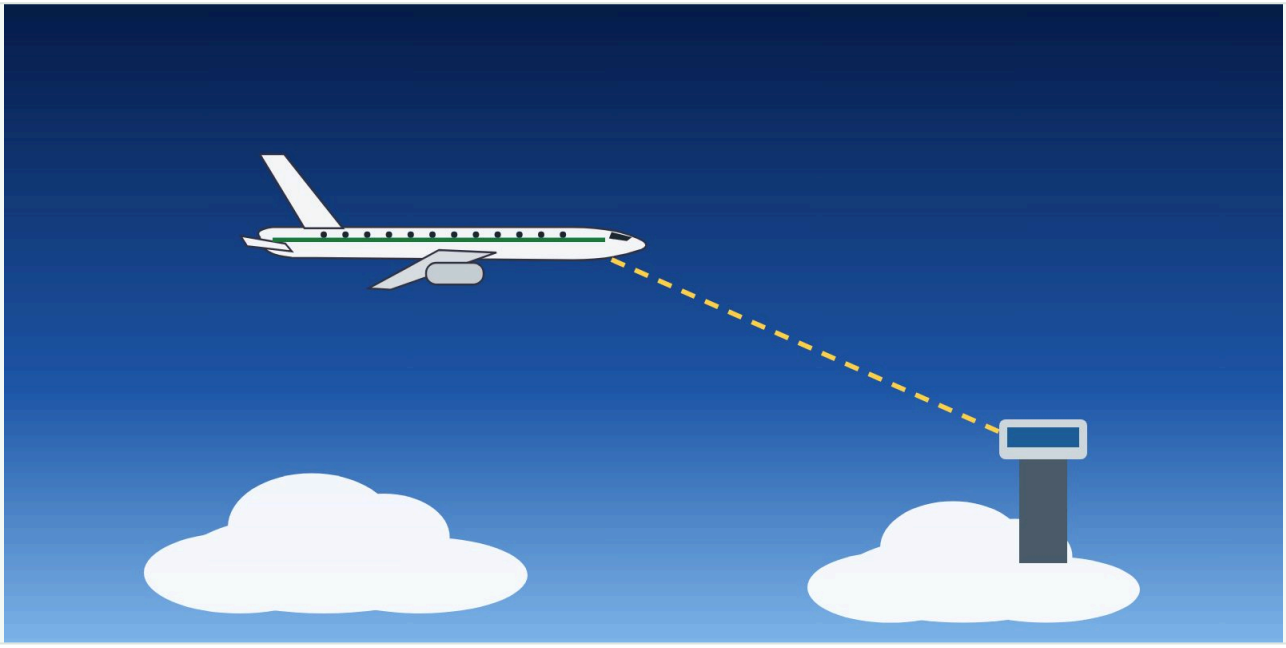
MODULE 1 · CPDLC PRINCIPLES

Lesson 01: CPDLC: Definition, References and Benefits

Lesson 1 of 6 · about 7 min

LEARNING OBJECTIVES

- Define CPDLC and describe its purpose.
- Identify the ICAO and European references.
- List the expected benefits and limitations.



CPDLC: data link messages between controller and pilot.

Definition

Controller–Pilot Data Link Communications (CPDLC) is a means of communication between controller and pilot using data link for ATC communications (ICAO Doc 4444, PANS-ATM). It is a two-way system by which controllers can send **non-urgent, strategic messages** to an aircraft as an alternative to voice. The message is displayed on a flight deck display.

- Controllers can issue clearances (levels, routes and deviations, speeds), frequency assignments and requests for information.
- Pilots can respond to messages, request and receive clearances and information, and send reports.
- A **free text** capability exchanges information that does not fit the standard formats.

References

Reference	Content
ICAO Annex 10, Volume III	Communication systems, including data link

ICAO Doc 4444 (PANS-ATM)	CPDLC procedures and the CPDLC message set
ICAO Doc 10037 — GOLD (Global Operational Data Link) Manual	The global operational guidance for flight crews, operators and ATS on data link operations
Commission Implementing Regulation (EU) 2023/1770	Current EU requirements for aircraft equipment and data link services in Single European Sky airspace. It **repealed** the original Data Link Services Implementing Rule, Regulation (EC) No 29/2009.
Aircraft manuals	FCOM / AFM limitations and the CMU operating procedures for your type

A short history in Europe

The Data Link Services Implementing Rule (Regulation (EC) No 29/2009) mandated CPDLC above FL285 in European airspace. Its implementation was affected by technical problems — random disconnections called **provider aborts**. An EASA investigation found that they came from a combination of factors linked to the radio-frequency environment and the single-frequency data link infrastructure, and recommended a multi-frequency infrastructure. The dates of application were therefore deferred (Regulation (EU) 2015/310 and later amendments). In 2023 the requirements were consolidated in Regulation (EU) 2023/1770.

Benefits

- Less voice traffic on the frequency and increased sector capacity;
- more pilot requests handled at the same time;
- fewer misunderstandings — for example call sign confusion;
- safer frequency changes and fewer loss-of-communication events.

LIMITATION

CPDLC latency is much longer than voice — a response can take up to several minutes. Use it only for **non-time-critical** communications. Check your aircraft limitations: on the B737 with protected-mode CPDLC (PM-CPDLC), the installation is intended for use in the **cruise** phase and for non-critical messages only.

KEY TAKEAWAYS

- CPDLC: data link ATC messages for non-urgent, strategic communications.
- References: Annex 10 Vol III, PANS-ATM, ICAO GOLD (Doc 10037), EU Regulation 2023/1770.
- Benefits: less frequency congestion, fewer misunderstandings; limitation: latency — not for time-critical messages.

KNOWLEDGE CHECK

Which regulation now contains the EU data link requirements, replacing Regulation (EC) No 29/2009?

A. Regulation (EU) 965/2012

B. Commission Implementing Regulation (EU) 2023/1770

C. Regulation (EU) 2015/340

D. ICAO Doc 4444

Answer: B. Regulation (EU) 2023/1770 repealed Regulation (EC) No 29/2009.

MODULE 1 · CPDLC PRINCIPLES

Lesson 02: Data Link Services

Lesson 2 of 6 · about 6 min

LEARNING OBJECTIVES

- Describe the data link services available to the flight crew.
- Explain the logon (DLIC) process.
- Identify the services mandated in Europe.



Log on before entering CPDLC airspace.

The services

Service	Purpose
DLIC — Data Link Initiation Capability	Provides the information needed to establish data link communications between an ATS unit and the aircraft. It is done **before any other data link service** (the logon).
ACM — ATC Communications Management	Automated assistance for the transfer of ATC communications (voice and CPDLC) between sectors and units.
ACL — ATC Clearances	Operational exchanges: pilots send requests and reports; controllers issue clearances, instructions and notifications.
AMC — ATC Microphone Check	Lets the controller send one instruction to all CPDLC aircraft on a frequency to check that none is blocking the voice channel (stuck microphone).
DCL — Departure Clearance	Requesting and delivering departure clearances by data link.
DSC — Downstream Clearance	Obtaining a clearance from an ATS unit that is not yet in control of the aircraft, when it cannot be obtained through unit-to-unit coordination.

The European data link rules mandated the first four services: **DLIC, ACM, ACL and AMC**.

Logon

- The logon uses the **ICAO code of the ATS unit** and must match the flight plan data (aircraft identification and registration). Errors in the flight plan or the FMS cause logon failures.
- Log on in good time before entering the CPDLC airspace, as published in the AIP.
- Only one ATS unit is the **current data authority** at any time — a controlled flight is under the control of only one ATC unit at a time (ICAO Annex 11).

KEY TAKEAWAYS

- DLIC (logon) comes first; then ACM, ACL and AMC — the four mandated services in Europe.
- DCL and DSC add departure and downstream clearances.
- Logon details must match the flight plan; only one current data authority at a time.

KNOWLEDGE CHECK

What does the AMC service do?

A. Sends weather reports

B. Lets the controller check that no aircraft is blocking the voice frequency

C. Assigns squawk codes

D. Manages the logon

Answer: B. The ATC Microphone Check service detects a stuck microphone blocking the frequency.

MODULE 1 · CPDLC PRINCIPLES

Lesson 03: Principles of CPDLC Exchanges and Message Composition

Lesson 3 of 6 · about 7 min

LEARNING OBJECTIVES

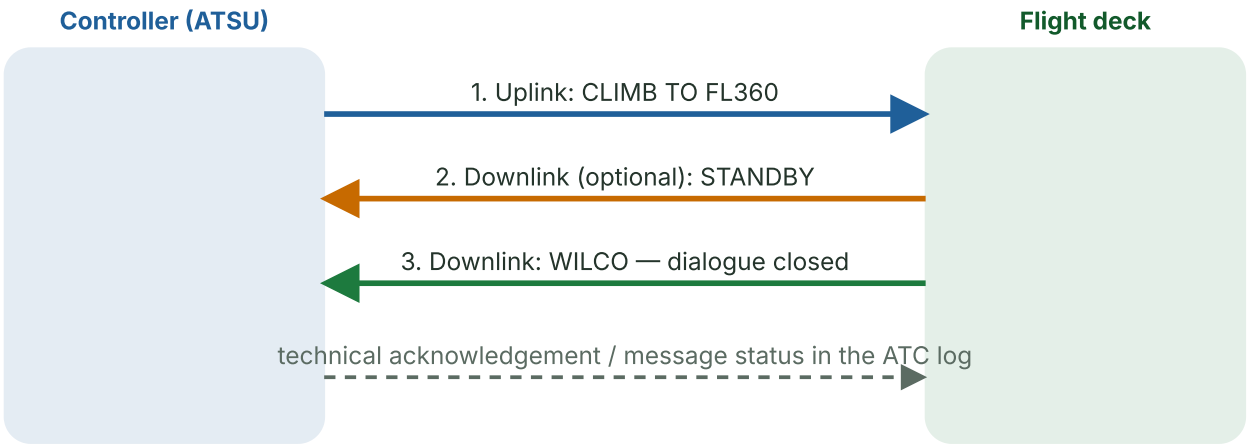
- Apply the principles for choosing voice or CPDLC.
- Understand single- and multi-element messages.
- Respond correctly to messages.



Read the whole message before responding.

Principles

- **Voice and data link co-exist.** CPDLC supplements voice; it does not replace it.
- CPDLC is used only for **non-time-critical** communications. Time-criticality depends on the traffic situation, the end-to-end performance (systems and human response times) and the recovery time.
- The choice of voice or CPDLC is at the discretion of the controller and the pilot.
- Usually, a CPDLC message is answered by CPDLC and a voice message by voice.



Read the whole message, cross-check with the other pilot, then respond — and only then act.

A typical CPDLC dialogue. The response closes the dialogue; STANDBY keeps it open.

Message composition

- Messages use a **standard format** from the ICAO message set, in plain language or abbreviations — without non-essential words.
- A message can have a **single element** (for example `CLIMB TO FL360`) or **several elements** (for example `CLIMB TO FL360, CLIMB AT 1000 FT PER MINUTE MINIMUM`). The maximum number of elements depends on the system and the airspace.
- A single response to a multi-element message applies to **all elements**. If any part cannot be complied with, respond **UNABLE** to the whole message.
- Only one open dialogue of the same type with the same aircraft is allowed at a time: horizontal profile, vertical profile, speed, or SSR code. A speed and a level dialogue can be open together; a heading and a direct-route dialogue cannot.
- Systems must handle unsupported messages correctly, normally with an error message.

Responses

Response	Meaning
WILCO	I will comply with the clearance or instruction.
UNABLE	I cannot comply (with the whole message).
STANDBY	Wait — the message is being considered; a further response will follow. The dialogue stays open.
AFFIRM / NEGATIVE	Yes / no to a question.
ROGER	Message received (for information messages).

KEY TAKEAWAYS

- Voice and CPDLC co-exist; CPDLC for non-time-critical messages only.
- One response applies to all elements of a multi-element message — UNABLE if any part cannot be done.
- Only one open dialogue of each type at a time.

KNOWLEDGE CHECK

You receive a two-element message and can comply with only one element. What do you respond?

A. WILCO

B. UNABLE for the whole message

C. ROGER

D. AFFIRM

Answer: B. If any element cannot be complied with, respond UNABLE to the whole message.

MODULE 2 · PROCEDURES AND PRACTICE

Lesson 04: Reverting to Voice, Phraseology and Failures

Lesson 4 of 6 · about 6 min

LEARNING OBJECTIVES

- Recognise when to revert from CPDLC to voice.
- Use CPDLC-related voice phraseology.
- Handle message failures and CPDLC system failures.



When in doubt, revert to voice.

When to revert to voice

- To clarify an unexpected, inappropriate or ambiguous CPDLC message.
- To make sure an instruction is executed in time.
- To correct an unintended message that has been sent.
- When a time-out or an error occurs for a CPDLC message.
- Always for urgent or emergency situations.

Phraseology

Situation	Phraseology
Correcting a CPDLC message by voice	"(call sign) DISREGARD CPDLC (message content or type) MESSAGE, BREAK, (corrected message)" Example: "DOMESTIC 3012, DISREGARD CPDLC CLIMB MESSAGE, BREAK, CLIMB FLIGHT LEVEL 360."
Failure of a single message	Confirm by voice the action that will be taken, prefaced with "CPDLC MESSAGE FAILURE" — or resend the message by CPDLC.

CPDLC has failed	Revert to voice and inform the other party: “CPDLC FAILURE”.
Complete ground system failure	Controller: “ALL STATIONS CPDLC FAILURE (unit name)”.
Stop sending requests	“((call sign) or ALL STATIONS) STOP SENDING CPDLC REQUESTS [UNTIL ADVISED] [(reason)]”
Resume normal use	“((call sign) or ALL STATIONS) RESUME NORMAL CPDLC OPERATIONS”

GOOD PRACTICE

Keep listening to the voice frequency at all times while using CPDLC. A clearance received by voice takes precedence over a pending CPDLC message on the same subject — clarify if in doubt.

KEY TAKEAWAYS

- Revert to voice to clarify, to ensure timely execution, to correct, after a time-out or error, and always in an emergency.
- Use “DISREGARD CPDLC … MESSAGE, BREAK …”, “CPDLC MESSAGE FAILURE”, “CPDLC FAILURE”.
- Keep a listening watch on voice.

KNOWLEDGE CHECK

A CPDLC message times out without a response. What should you do?

A. Send it again repeatedly

B. Revert to voice to confirm the clearance

C. Ignore it

D. Switch off the CMU

Answer: B. After a time-out or error, revert to voice.

MODULE 2 · PROCEDURES AND PRACTICE

Lesson 05: Using the CMU-900 on the Boeing 737-800

Lesson 5 of 6 · about 7 min

LEARNING OBJECTIVES

- Navigate the main CPDLC pages of the CMU-900 on the CDU.
- Respond to uplinked messages.
- Use the ATC log and the reports and requests pages.



On the B737-800, CPDLC runs through the CMU-900 on the CDU.

The application

On our Boeing 737-800, the **CMU-900** CPDLC application allows digital communication between the aircraft and an ATC centre through the Aeronautical Telecommunication Network (ATN), using the CDU.

- **LOGON/STATUS** page — initiates the logon to a specified ATS facility and displays the data link status.
- **REPORTS/REQUESTS** menu — creates downlink requests: altitude, speed or direct (REQ ALT/SPD/DIR), weather deviation (REQ WX DEV) and facility monitoring (MONITORING).
- **ATC LOG** — lists all uplinks and downlinks sent and received.

Receiving a message

- When an uplink is received, a **Hi/Lo chime** sounds and **•ATC MSG** appears at LSK 6R on the CDU. The prompt remains while any uplink is pending.
- Open the message, read it **in full**, and cross-check it with the other pilot.

- Respond with the appropriate prompt: **WILCO, STANDBY, AFFIRM, UNABLE, ROGER or NEGATIVE**.
- Selecting **STANDBY** sends a downlink and restarts the alert timer; the message is not closed until another response is sent. Selecting ROGER after STANDBY replaces STANDBY with ACCEPTED, and the message is completed.

ATC log

- Shows, for each message: uplink or downlink, time, source, tracking number, status and the first 22 characters of the text.
- Select the message text to display the complete message with its response.
- Status annunciations show whether a message is open, accepted, rejected, failed or timed out — check them when in doubt.

Dash 8 fleets

Q400 and Q200 crews: check whether your aircraft is CPDLC equipped. If not, this course gives awareness of CPDLC operations used by other traffic and ATC.

Button names and page layouts depend on the installed software — always use your FCOM.

KEY TAKEAWAYS

- CMU-900 pages: LOGON/STATUS, REPORTS/REQUESTS, ATC LOG.
- Uplink: chime + •ATC MSG at LSK 6R; read in full, cross-check, then respond.
- STANDBY keeps the dialogue open; check the status annunciations in the ATC log.

KNOWLEDGE CHECK

On the CMU-900, how is the receipt of an uplink message indicated?

A. Master caution only

B. A Hi/Lo chime and •ATC MSG at LSK 6R on the CDU

C. A message on the PFD only

D. No indication

Answer: B. A Hi/Lo chime sounds and •ATC MSG is displayed at LSK 6R.

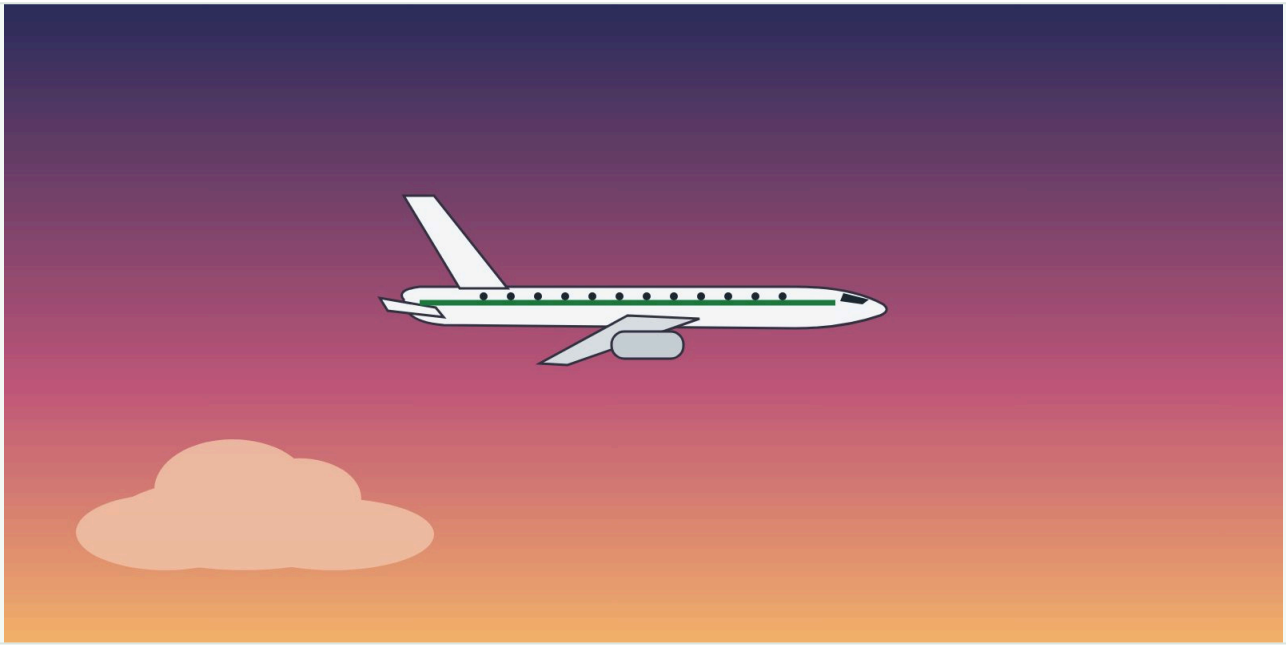
MODULE 2 · PROCEDURES AND PRACTICE

Lesson 06: Good Practice and Common Errors

Lesson 6 of 6 · about 5 min

LEARNING OBJECTIVES

- Apply crew procedures that prevent CPDLC errors.
- Recognise common CPDLC errors.
- Manage workload when using CPDLC.



Only one pilot heads-down at any time.

Crew procedures

1. **Read the whole message** — aloud or silently, as the SOP requires. Conditional clauses ("AT … CLIMB TO …", "AFTER PASSING …") are easy to miss.
2. **Cross-check** it with the other pilot before responding.
3. **Respond** — then set the clearance in the FCU/MCP or FMS, and cross-check the result.
4. If anything is unclear, **revert to voice**.

Common errors

- Executing a message before responding, or responding WILCO without both pilots reading it.
- Missing a condition in a conditional clearance.
- Responding to the wrong message when several are pending.
- Logon failures caused by wrong flight plan or FMS data.
- Heads-down work with CPDLC at a critical moment — only one pilot should be heads-down at any time.

WORKLOAD

CPDLC reduces frequency congestion, but it can increase heads-down time. Plan its use for cruise and for non-urgent exchanges, and keep the pilot flying focused on the flight path.

KEY TAKEAWAYS

- Read in full, cross-check, respond, then act — and verify the result.
- Watch for conditional clauses and multiple pending messages.
- Only one pilot heads-down; revert to voice when in doubt.

KNOWLEDGE CHECK

What is the correct order when handling a CPDLC clearance?

A. Act, then read, then respond

B. Read in full, cross-check, respond, then act and verify

C. Respond, then read

D. Act without responding

Answer: B. Read, cross-check, respond, then act and verify.

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